

KALACHINSK, Russian Federation

WMO: 286960

Lat: 55.0193N

Lon: 74.6125E

Elev: 362

StdP: 14.50

Time Zone: 6.00 (E06)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.7	-25.4	-36.6	0.7	-30.7	-31.3	1.0	-25.2	22.8	15.2	20.5	15.7	3.2	30	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	86.9	66.2	83.7	65.0	80.4	63.8	69.3	81.5	67.6	78.7	65.9	76.5	9.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	94.8	73.7	63.5	88.9	72.2	61.8	83.7	70.9	33.6	81.6	32.2	78.8	30.9	76.7	91.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.3	18.9	16.5	-34.5	91.4	8.1	3.9	-40.3	94.2	-45.1	96.5	-49.6	98.7	-55.5	101.5		
			-34.7	71.7	8.1	1.8	-40.5	73.0	-45.2	74.0	-49.7	75.0	-55.6	76.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.2	-1.1	3.1	19.3	39.0	54.0	63.6	66.2	61.9	50.8	37.9	20.0	5.2	(6)
(7)		DBStd	26.18	14.53	13.06	10.65	11.13	9.71	7.86	6.69	7.35	8.50	8.72	13.43	14.60	(7)
(8)		HDD50	7051	1583	1314	953	360	70	5	0	3	91	384	900	1388	(8)
(9)		HDD65	11154	2048	1734	1418	781	360	117	69	149	435	840	1350	1853	(9)
(10)		CDD50	1634	0	0	0	29	195	414	501	372	114	9	0	0	(10)
(11)		CDD65	261	0	0	0	0	19	75	106	53	8	0	0	0	(11)
(12)		CDH74	2789	0	0	0	11	298	807	1039	544	90	0	0	0	(12)
(13)		CDH80	799	0	0	0	1	69	238	312	163	16	0	0	0	(13)
(14)	Wind	WSAvg	7.1	6.2	6.5	7.5	8.5	8.2	6.7	6.4	6.3	6.8	7.2	7.7	7.2	(14)
(15)	Precipitation	PrecAvg	14.2	0.6	0.6	0.6	0.8	1.1	2.0	2.4	2.0	1.1	1.1	1.1	0.9	(15)
(16)		PrecMax	21.6	1.4	1.5	1.5	2.0	3.1	4.5	7.0	5.0	2.6	2.9	2.3	1.8	(16)
(17)		PrecMin	9.5	0.0	0.1	0.0	0.0	0.2	0.6	0.1	0.6	0.3	0.4	0.2	0.3	(17)
(18)		PrecStd	2.9	0.4	0.3	0.4	0.5	0.6	0.9	1.4	0.9	0.7	0.6	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.8	34.1	44.3	74.4	86.6	89.9	91.4	90.9	81.9	65.6	45.2	34.8	(19)
MCWB			31.8	31.9	37.7	54.4	61.3	65.4	70.4	67.0	59.9	51.9	42.3	32.7	(20)	
2%		DB	28.6	30.1	38.9	66.9	81.3	85.9	86.7	84.5	76.0	60.1	41.7	32.6	(21)	
		MCWB	27.4	28.7	35.3	50.9	59.5	64.7	67.4	66.6	57.9	49.0	39.3	31.3	(22)	
5%		DB	24.7	27.0	35.9	62.0	77.2	82.4	83.3	79.6	69.7	55.1	38.7	28.9	(23)	
		MCWB	23.5	25.7	33.4	49.2	57.5	64.5	66.4	64.1	56.2	46.9	36.3	27.5	(24)	
10%		DB	21.0	22.9	33.8	56.6	72.2	78.4	79.9	75.9	65.2	51.1	35.0	24.9	(25)	
		MCWB	20.1	21.7	31.7	46.4	55.4	63.2	65.6	62.5	53.7	44.7	32.9	23.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.9	32.1	38.3	56.5	64.3	70.3	72.9	70.5	63.7	53.9	43.3	33.3	(27)
MCD B			32.9	33.5	42.9	69.3	81.1	83.1	81.3	83.8	78.4	61.4	44.8	34.0	(28)	
2%		WB	27.3	28.8	35.7	52.7	61.2	68.0	70.0	67.9	59.9	51.0	39.7	31.2	(29)	
		MCD B	28.4	30.1	38.1	65.0	77.9	79.4	82.1	79.5	71.5	57.5	41.7	32.5	(30)	
5%		WB	23.7	25.7	33.8	49.7	58.8	66.0	68.3	65.9	57.2	47.7	36.3	27.7	(31)	
		MCD B	24.8	27.0	35.7	61.0	73.6	77.9	79.6	76.3	67.2	54.0	38.5	29.1	(32)	
10%		WB	20.0	21.6	31.7	46.8	56.5	64.2	66.7	64.0	54.6	45.0	33.2	23.6	(33)	
		MCD B	21.0	23.0	33.7	56.2	69.8	75.8	77.1	73.4	64.0	50.6	34.7	24.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.2	14.4	14.6	16.8	21.5	19.9	18.9	19.0	17.5	13.3	10.7	13.0	(35)
MCDBR			15.8	12.5	12.5	25.7	27.7	25.7	24.6	25.2	25.2	21.5	11.2	13.7	(36)	
MCWBR			15.6	12.1	10.2	13.9	12.3	10.5	9.5	11.0	12.0	13.0	9.9	13.7	(37)	
5% WB		MCD BR	15.2	12.7	11.5	23.6	25.0	21.8	21.0	21.6	22.0	18.7	11.0	14.1	(38)	
		MCWBR	15.0	12.4	10.1	13.5	12.5	10.4	9.9	11.0	11.3	12.3	9.9	14.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.284	0.316	0.376	0.405	0.401	0.410	0.432	0.429	0.368	0.353	0.324	0.288	(40)	
(41)		taud	2.043	2.014	1.926	2.127	2.228	2.285	2.216	2.191	2.322	2.231	2.095	2.000	(41)	
(42)		Ebn at Noon	193	231	242	257	267	266	256	248	250	219	174	157	(42)	
(43)		Edh at Noon	22	32	45	43	41	39	41	40	30	26	21	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	311	656	1159	1473	1766	1905	1748	1408	1007	515	246	201	(44)	
(45)		RadStd	22	29	79	115	164	135	117	117	91	65	36	23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page